Project: Worked examples
Title: Tolerance stack-up (1D) v2
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

A linear tolerance loop: worst-case limits, root-sum-square and modified RSS limits, predicted spread and capability against a stated requirement, contribution Pareto and a Monte Carlo prediction.
Worst case: Σ|half-bands|. RSS: √Σ half². Modified RSS: factor × RSS (Bender-style). Predicted σ: √Σ σ_i² with σ_i from each contributor's distribution. C_pk from the nearer limit. Monte Carlo: seeded sampling of every contributor, summed with its sign.
Ref: BS 8888 — Technical product documentation and specification
Ref: ASME Y14.5 — Dimensioning and tolerancing
Ref: ISO 2768-1 — General tolerances (where a contributor's band is a general tolerance)

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
contributors	Contributors in the loop	3 row(s)		drawing: DRG-HSG-001 rev B	any
contributors[1]		name = Housing bore depth; direction = adds; nominal = 40 mm; upper = 0.1 mm; lower = 0 mm			
contributors[2]		name = Bearing width; direction = subtracts; nominal = 20 mm; upper = 0 mm; lower = -0.12 mm			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
contributors[3]		name = Spacer; direction = subtracts; nominal = 19.5 mm; upper = 0.02 mm; lower = -0.02 mm			
minimum	Minimum acceptable result	0.4	mm	spec: SPEC-BRG-7 §4.2	-1000000 … 1000000 mm
maximum	Maximum acceptable result	0.8	mm	spec: SPEC-BRG-7 §4.2	-1000000 … 1000000 mm
k	Standard deviations per half-band (normal contributors)	3		assumption: Each part at ±3σ within its band.	1 … 6
f_rss	Modified-RSS factor (Bender-style)	1.5		assumption: Bender factor 1.5.	1 … 2
cpk_target	Required C_pk	1.33		assumption: Usual capability target.	1 … 2
n_mc	Monte Carlo samples	10000		assumption: Default sample count.	1000 … 100000
seed	Monte Carlo seed	1		assumption: Fixed seed for reproducibility.	0 … 4295000000
notation	Drawing notation	BS 8888		assumption: Metric drawing.	any

3. Assumptions

A1. Every contributor acts along the one axis of the stack. (A linear stack: no angular or datum-shift effects beyond what the user enters as a contributor.)
A2. Statistical figures assume contributors vary independently about their mid-limit (plus any stated mean shift), each with the distribution stated for it. (The worst case is a guarantee; every statistical figure is a prediction about a population.)
A3. A position or profile contributor contributes ± half its tolerance zone, plus half the stated MMC bonus. (Diametral zones are bilateral about the true position.)
A4. k (Standard deviations per half-band (normal contributors)) is taken as 3. (Each part at ±3σ within its band.)
A5. f_rss (Modified-RSS factor (Bender-style)) is taken as 1.5. (Bender factor 1.5.)
A6. cpk_target (Required C_pk) is taken as 1.33. (Usual capability target.)
A7. n_mc (Monte Carlo samples) is taken as 10000. (Default sample count.)
A8. seed (Monte Carlo seed) is taken as 1. (Fixed seed for reproducibility.)
A9. notation (Drawing notation) is taken as BS 8888. (Metric drawing.)

4. Working

[7] c1 = + 40 +0.1/0
= mid 40.05 mm, ± 0.05 mm, □ 0.0166667 mm (normal)
= 40.05 mm
Housing bore depth
[8] c2 = - 20 +0/-0.12
= mid 19.94 mm, ± 0.06 mm, □ 0.02 mm (normal)
= -19.94 mm
Bearing width
[9] c3 = - 19.5 ±0.02
= mid 19.5 mm, ± 0.02 mm, □ 0.00666667 mm (normal)
= -19.5 mm
Spacer
[10] x_nom = □ (sign × nominal)
= 40 - 20 - 19.5
= 0.5 mm
Nominal result
[11] x_mean = □ (sign × mid-limit)
= 40.05 - 19.94 - 19.5
= 0.61 mm
Mean result
[12] t_wc = □ half-bands
= 0.05 + 0.06 + 0.02
= 0.13 mm
Worst-case tolerance (±)
[13] x_wc_min = x_mean - t_wc
= 0.61 mm - 0.13 mm
= 0.48 mm
Worst-case minimum
[14] x_wc_max = x_mean + t_wc
= 0.61 mm + 0.13 mm
= 0.74 mm
Worst-case maximum
[15] t_rss = √□ half²
= √(0.05² + 0.06² + 0.02²)
= 0.080623 mm
Root-sum-square tolerance (±)
[16] x_rss_min = x_mean - t_rss
= 0.61 mm - 0.080623 mm
= 0.52938 mm
RSS minimum
[17] x_rss_max = x_mean + t_rss
= 0.61 mm + 0.080623 mm
= 0.69062 mm
RSS maximum
[18] t_mrss = f_rss * t_rss
= 1.5 × 0.080623 mm
= 0.12093 mm
Modified RSS tolerance (±)
[19] sigma = √□ □_i²
= √(0.0166667² + 0.02² + 0.00666667²)
= 0.026874 mm
Predicted standard deviation of the result

[20] m_min = x_wc_min - minimum
= 0.48 - 0.4
= 0.08 mm
Worst-case margin to the minimum
[21] m_max = maximum - x_wc_max
= 0.8 - 0.74
= 0.06 mm
Worst-case margin to the maximum
[22] cpk = min(x_mean - LSL, USL - x_mean) / (3 ⊠)
= min(0.21, 0.19) / (3 × 0.0268742)
= 2.3567
Process capability index C_pk
[23] cp = (USL - LSL) / (6 ⊠)
= (0.8 - 0.4) / (6 × 0.0268742)
= 2.4807
Process capability C_p
[24] dpmo = 10⁶ × P(outside)
= 10⁶ × ⊠-tails at ⊠ = 0.0268742
= 7.7755e-7
Predicted defects per million
[25] yield = 1 - dpmo / 10⁶
= 1 - 7.7755e-7 / 10⁶
= 100 %
Predicted yield
[26] u_cpk = cpk_target / cpk
= 1.33 / 2.3567
= 0.56436
Capability utilisation
[27] pareto = largest contributor (statistical)
= Housing bore depth 38.5 %; Bearing width 55.4 %; Spacer 6.15 %
= "Bearing width"
Contribution Pareto
[30] mc_mean = mean of 10000 samples (seed 1)
= Monte Carlo
= 0.60987 mm
Monte Carlo mean
[31] mc_sigma = sample standard deviation
= Monte Carlo
= 0.027228 mm
Monte Carlo σ
[32] mc_p05 = 5th percentile
= Monte Carlo
= 0.56589 mm
Monte Carlo 5 %
[33] mc_p95 = 95th percentile
= Monte Carlo
= 0.65462 mm
Monte Carlo 95 %
[34] mc_yield = samples inside the requirement
= 10000 of 10000
= 100 %
Monte Carlo yield

5. Results

SYMBOL	VALUE	DESCRIPTION
x_nom	0.5 mm	Nominal result
x_mean	0.61 mm	Mean result
t_wc	0.13 mm	Worst-case tolerance (±)
x_wc_min	0.48 mm	Worst-case minimum
x_wc_max	0.74 mm	Worst-case maximum
t_rss	0.080623 mm	RSS tolerance (±)
t_mrss	0.12093 mm	Modified RSS tolerance (±)
sigma	0.026874 mm	Predicted σ
m_min	0.08 mm	Worst-case margin to the minimum
m_max	0.06 mm	Worst-case margin to the maximum
cp	2.4807	C_p
cpk	2.3567	C_pk
dpmo	7.7755e-7	DPMO (predicted)
yield	100 %	Yield (predicted)
mc_mean	0.60987 mm	Monte Carlo mean
mc_sigma	0.027228 mm	Monte Carlo σ
mc_p05	0.56589 mm	Monte Carlo 5th percentile
mc_p95	0.65462 mm	Monte Carlo 95th percentile
mc_yield	100 %	Monte Carlo yield

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	C_pk against target	u_cpk = 0.56436	u_cpk ≤ 1	0.564	
PASS	Worst case within requirement	0.48 … 0.74 mm	0.4 … 0.8 mm	0	

7. Applicability

WARNING — few-contributors: contributors: fewer than four contributors; RSS and modified RSS are optimistic for so few.

8. Verification

Template mech.tolerance-stack@2.0.0 Content hash 2e12e5e8f8c8d342 Engine calc-engine/1.0.0
Known-answer cases passed 6/6 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0